AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 26 · STUDENT LESSON

# Incident investigation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

## Record what happened. Separate what you infer. Test what could explain it.

Use minimized timelines and controlled fictional tests without inventing a root cause.

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**Keep one habit:** Evidence contract → morning timeline → afternoon trace picture → hypothesis probes → candidate fixture → independent permission incident.

MAIN LESSON · SECTION 1

## Build an evidence timeline before naming the exact cause

> Separate observations, candidate explanations, containment and controlled tests.

### Learning goals

I can explain the **mechanism** and terminology of incident investigation.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

## Retrieve monitoring, versions and cache boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Monitoring quality drift. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

## Prerequisite and starting response

- Recall Monitoring quality drift. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

## End goal

Explain incident investigation using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

**Your learning target**

**Remember before reading**

This week you will practise:

Explain the **mechanism** and terminology of incident investigation

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

**Retrieve what you already know**

**This week's end goal**

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MAIN LESSON · SECTION 2

# Use observations to constrain hypotheses

> Preserve permitted evidence while separating containment from diagnosis.

## Begin with the changed school noticeboard

Imagine the school noticeboard changes the classroom, but a helper still gives the old room. First write what was observed and when. "The model changed" is a hypothesis, not an observation. Compare version records and the evidence actually used.

Prompt containment can proceed while an authorized person captures minimized version and event metadata; do not delay necessary containment just to collect a perfect record, or copy private payloads into an investigation notebook.

## Define an incident timeline

Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

### Complete fictional evidence and test contract

| Evidence or action | Complete classroom meaning |
| --- | --- |
| Observation | Supplied timestamp, version, trace field or outcome; preserve source and uncertainty |

| Evidence or action | Complete classroom meaning |
|---|---|
| Hypothesis | Proposed explanation that must be compared with evidence |
| Main trusted version record | Weight version w7 before and at response; c2 published at 09:00, response uses c1 at 09:05 |
| Containment | Cache disabled at 09:06 in the supplied incident record; limits exposure, does not establish cause |
| Candidate controlled test | Same query/source/weight fixture; vary only whether buggy query-only cache is used |
| Permitted investigation record | Fictional IDs, times, versions, actions and necessary outcome summaries; no unnecessary private content |
| Picture evidence localization | Old retrieval version versus current retrieval followed by wrong draft are different mismatches |
| Limits | Reliable declared metadata assumed; no live reproduction, complete timeline guarantee or causal proof from temporal order |

## Name supplied and missing evidence

**List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

# Trace event evidence and the claim each row can support

> A plausible explanation must survive the supplied version record.

## Original minimized timeline procedure

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**.

### Complete main event and inference ledger

| Main supplied event | Observation | What it supports and does not prove |
|---|---|---|
| Before response | Weight version w7 | Baseline under reliable supplied version metadata |
| 09:00 | Corpus c2 published | New source predates the stale response |
| 09:05 | Response uses old c1; weight version still w7 | Stale evidence observed; reported weight-change explanation conflicts with declared metadata |
| 09:06 | Cache disabled | Containment action; no supplied later success rate |
| Added controlled fixture | Cache enabled gives c1; disabled gives c2 | A buggy query-only cache can reproduce this pattern locally; exact real bug remains unproven |

## State missing evidence before diagnosis

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

**Incident investigation — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against evidence. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

**Connect the picture and the explanation**

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MAIN LESSON · SECTION 4

# Predict a different trace with current evidence but the wrong draft

> Change the investigation target when the evidence changes.

## Current task and response

- At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

## Predict the changed trace

- Change: Trace instead shows version 6 retrieved and room C drafted.

▼ **Check after predicting**

**Prediction**: Investigate drafting or validation before blaming caching.

Reason: The mismatch now appears after retrieval.

▼ **Earlier wording, alternatives and picture transcripts**

**Compare a familiar case with a changed case**

**Change one thing in the pictured case**

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MAIN LESSON · SECTION 6

# Compare stale retrieval with a wrong draft from current evidence

> The trace locates the mismatch before it explains the mechanism.

## Main morning incident and supported candidate

At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged.

Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized **evidence** without unnecessary private content so a later revision can be compared honestly.

## Separate afternoon timeline and changed-trace scope

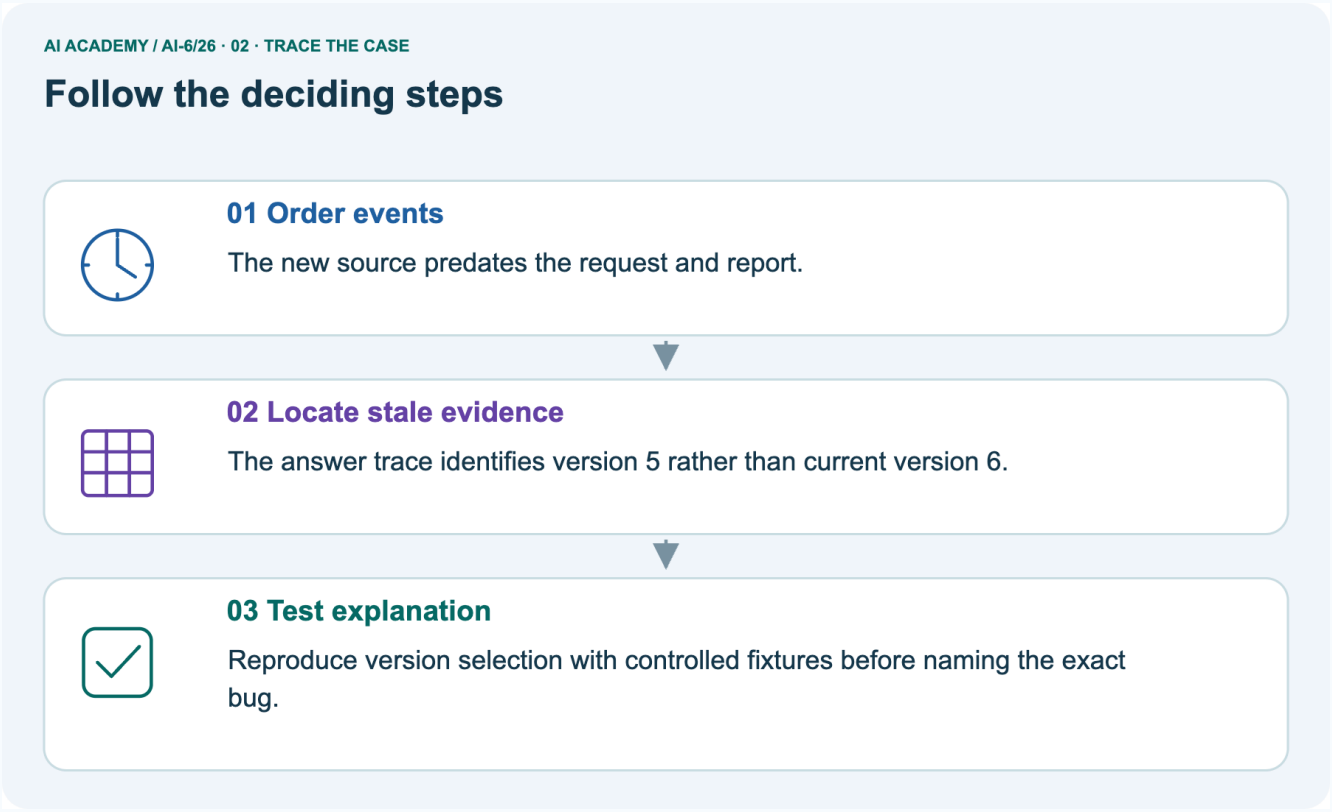
An **incident timeline**. After a schedule update, a service gives an old classroom location.

14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

The picture is a separate afternoon incident: 14:00 version 6 publishes room D; 14:05 the trace uses version 5 and drafts room C; 14:06 a user reports the mismatch. This locates stale evidence selection, but does not identify the exact code defect. Controlled version-selection/cache tests are appropriate.

In the changed trace, version 6 was retrieved but room C was drafted; now inspect context handoff, answer construction and output checking rather than asserting old-source retrieval. It is still possible for a cache elsewhere in a system to matter; this trace alone does not diagnose every component. Missing events or unreliable metadata may change the account.

## Trace events, evidence version and test selection




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Order events | The new source predates the request and report. |
| 2. Locate stale evidence | The answer trace identifies version 5 rather than current version 6. |
| 3. Test explanation | Reproduce version selection with controlled fixtures before naming the exact bug. |

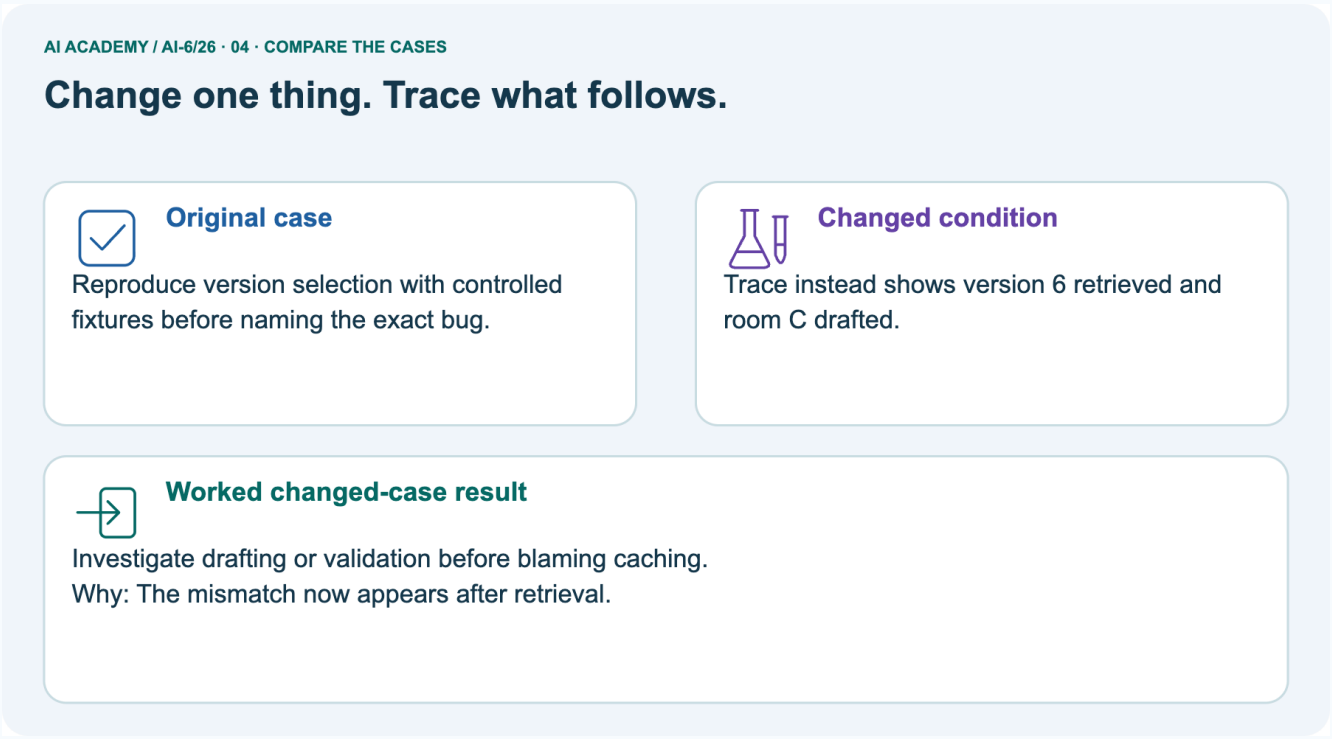
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened.

## Predict current version 6 with room C drafted

- Change one thing: Trace instead shows version 6 retrieved and room C drafted. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting




Follow the labelled picture with the matching explanation. Open the vector diagram

## Explain missing events and evidence localization

- Missing events may change the account; temporal order alone does not prove causation.
- Read the labelled picture: Order the three incident records and locate the old **evidence** version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?

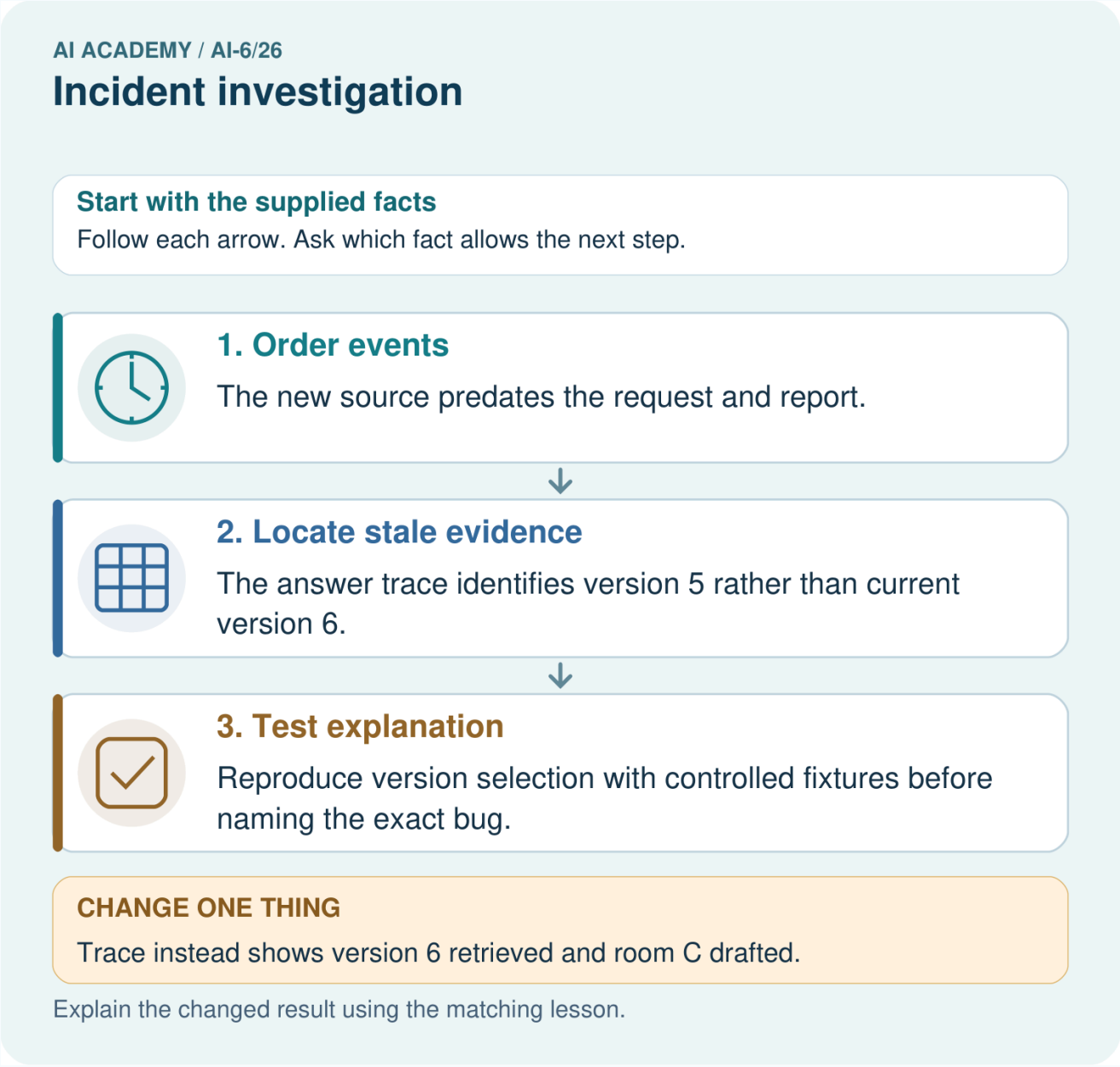
▼ Earlier wording, alternatives and picture transcripts

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Incident investigation

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

### Why this result follows

### What this example does not establish

MAIN LESSON · SECTION 7

# Audit the difference between localization and diagnosis

> Choose a test that could challenge the candidate explanation.

## Core incident qualification and failure analysis

A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

### Complete hypothesis, metadata and containment probes

| Investigation probe | Expected interpretation |
|---|---|
| Event B follows event A | Temporal order only; not sufficient causal proof |
| Trusted declared weight version unchanged | Contradicts a reported weight-version change under that metadata assumption |
| Old source version appears in trace | Evidence-selection mismatch; test version routing and cache candidates |
| Current source retrieved, incompatible room drafted | Inspect handoff/drafting/validation; do not label it old retrieval |
| Disabling cache recorded | Containment, not proof the incident is fully resolved |
| Candidate reproduced in a fictional harness | Supports possible mechanism in that harness; compare actual permitted incident evidence |
| Missing version or incomplete events | Mark evidence gap; do not invent it |

| Investigation probe | Expected interpretation |
| --- | --- |
| Necessary containment is urgent | Proceed promptly while collecting authorized minimized evidence where feasible |
| Private payload unnecessary to analysis | Exclude it; preserve permitted metadata instead |

## Guided repair and explanation

- A report omits this qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

**Locate the earliest unsupported step**

MAIN LESSON · SECTION 8

# Practise a timeline with explicit inference limits

> Label what you saw, what you think and what you would test.

## Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

## Supported W2 trace

Use workbook W2 for the connected example “Incident investigation”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Individual scope check

- Use the worked case for Incident investigation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

## Picture response

- Order the three incident records and locate the old **evidence** version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?

> **My explanation using a pictured step and the given facts**

## Reserve the selected permission-change incident

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

**Practise with less help: Trace the worked case**

**Check your reasoning**

**Explain the picture**

MAIN LESSON · SECTION 9

# Run the complete evidence-localization and candidate-test fixture

> Inspect what the experiment holds fixed and what conclusion it can support.

## Current morning-incident investigation

At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

## Runnable fictional incident model

```
"""Fictional incident evidence and controlled reproductions, no live log access."""
from copy import deepcopy

main_events=[{'time':'09:00','kind':'corpus_update','version':'c2'},

{'time':'09:05','kind':'response','source_version':'c1','weight_version':'w7'},
             {'time':'09:06','kind':'containment','action':'cache_disabled'}]
weight_before='w7';weight_at_response='w7'
before=deepcopy(main_events)
assert weight_before==weight_at_response
# A deliberately buggy query-only cache is one candidate mechanism, not the proven live
bug.
source={'c1':'old fictional answer','c2':'new fictional answer'}
cached={'query':'q','source_version':'c1','answer':source['c1']}
def candidate_response(cache_enabled):
 if type(cache_enabled) is not bool:raise ValueError('Boolean controlled condition
required')
 return deepcopy(cached) if cache_enabled else
{'query':'q','source_version':'c2','answer':source['c2']}
with_cache=candidate_response(True);without_cache=candidate_response(False)
assert with_cache['source_version']=='c1' and without_cache['source_version']=='c2'
assert main_events==before

def locate_mismatch(current_version,retrieved_version,drafted_room,records):
 if current_version not in records or retrieved_version not in records:
  return {'observed':'incomplete source evidence','next_test':'obtain permitted missing
```

```
version metadata'}
 if retrieved_version!=current_version:
  return {'observed':'stale evidence version','next_test':'reproduce version
selection/cache behavior'}
 if drafted_room!=records[retrieved_version]['room']:
  return {'observed':'draft differs from retrieved record','next_test':'inspect context
handoff and answer construction'}
 return {'observed':'no supplied version/room mismatch','next_test':'inspect other
reported symptoms'}
# Separate picture timeline and a changed trace; no model generates either draft.
records={'5':{'room':'C'},'6':{'room':'D'}}
picture_events=[('14:00','version 6 publishes room D'),('14:05','version 5 used, room C
drafted'),
                ('14:06','user reports mismatch')]
picture=locate_mismatch('6','5','C',records)
changed=locate_mismatch('6','6','C',records)
assert picture['observed']=='stale evidence version'
assert changed['observed']=='draft differs from retrieved record'
assert locate_mismatch('6','missing','C',records)['observed']=='incomplete source
evidence'
print('Main supplied timeline:',main_events)
print('Reliable declared weight version unchanged:',weight_before==weight_at_response)
print('Controlled candidate fixture, enabled/disabled:',with_cache,without_cache)
print('Picture ordered events:',picture_events)
print('Picture evidence localization:',picture,'changed trace:',changed)
print('Candidate reproduction supports a possible mechanism in this fixture; chronology
alone does not prove the real incident cause.')
print('Containment may proceed promptly while authorized minimized evidence is captured.
No private payloads, actual systems, model runs or live repairs are used.')
```

## Expected outcomes and interpretation limits

1. Predict which claim the unchanged w7 record challenges. The local controlled candidate keeps the query, source fixtures and declared weight context fixed while switching a deliberately buggy query-only cache on/off: old c1 versus current c2. This demonstrates one possible mechanism, not a discovered live root cause.

   The separate picture compares old evidence/old room with current evidence/wrong room and changes the recommended inspection accordingly. Inputs remain unchanged. No model generates these outputs, no logs or private records are read, and no actual cache or service is disabled. Mark paper traces “not run”.

## Evidence ledger and reflection

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

**Predict, investigate and preserve the evidence**

**Plan → try → check**

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MAIN LESSON · SECTION 14

# Try alone: investigate a permission change before a restricted read

> Propose a controlled test without exposing real restricted content.

### Complete selected independent task

10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

## First attempt and support record

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

**Independent case: Apply a new case independently**

**Attempt this without hints**

10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

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MAIN LESSON · SECTION 11

# Save the timeline, hypotheses and discriminating test plan

> Keep allowed metadata, original observations and revised explanations separate.

## Project task and artifact

Add a labelled incident investigation **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of incident investigation.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
|---|---|
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

## Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

**Evidence for your project**

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MAIN LESSON · SECTION 16

# Explain one supported inference and one unresolved cause

> Keep containment actions distinct from evidence of resolution.

## Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of incident investigation

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

## Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

## Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

### Vocabulary and your explanation

| Word | Meaning |
| --- | --- |
| Incident timeline | Ordered records of relevant events |
| Containment | An action limiting ongoing harm or failure |
| Root-cause hypothesis | A proposed mechanism requiring evidence |

**Use two terms to explain a step in this week's example.**

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

# Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

## Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

### Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

# Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

## Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

**My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

## Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

## Complete trace and evidence

### Complete evidence record

#### Incident investigation — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against evidence. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

# Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

## Targets,route and first prediction

Start with this question: An **incident timeline**. After a schedule update, a service gives an old classroom location.

| By the end, I can | How I will show it |
| --- | --- |
| Explain the mechanism and terminology of incident investigation | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- An **incident timeline**. After a schedule update, a service gives an old classroom location.
- Which labelled step matches this detail? Reproduce version selection with controlled fixtures before naming the exact bug.
- 1. Test explanation
- 2. Order events
- 3. Locate stale **evidence**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Test explanation Reproduce version selection with controlled fixtures before naming the exact bug. A timeline separates observed events from a proposed cause.

**Evidence** can locate a mismatch without yet explaining why it happened.

## Changed-case question and optional hint

- Changed-case practice: Trace instead shows version 6 retrieved and room C drafted. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Investigate drafting or validation before blaming caching.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

## Repair and retain the record

Review Monitoring quality drift (ai-6/25). Goal: Explain the **mechanism** and terminology of monitoring quality drift. Short practice: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

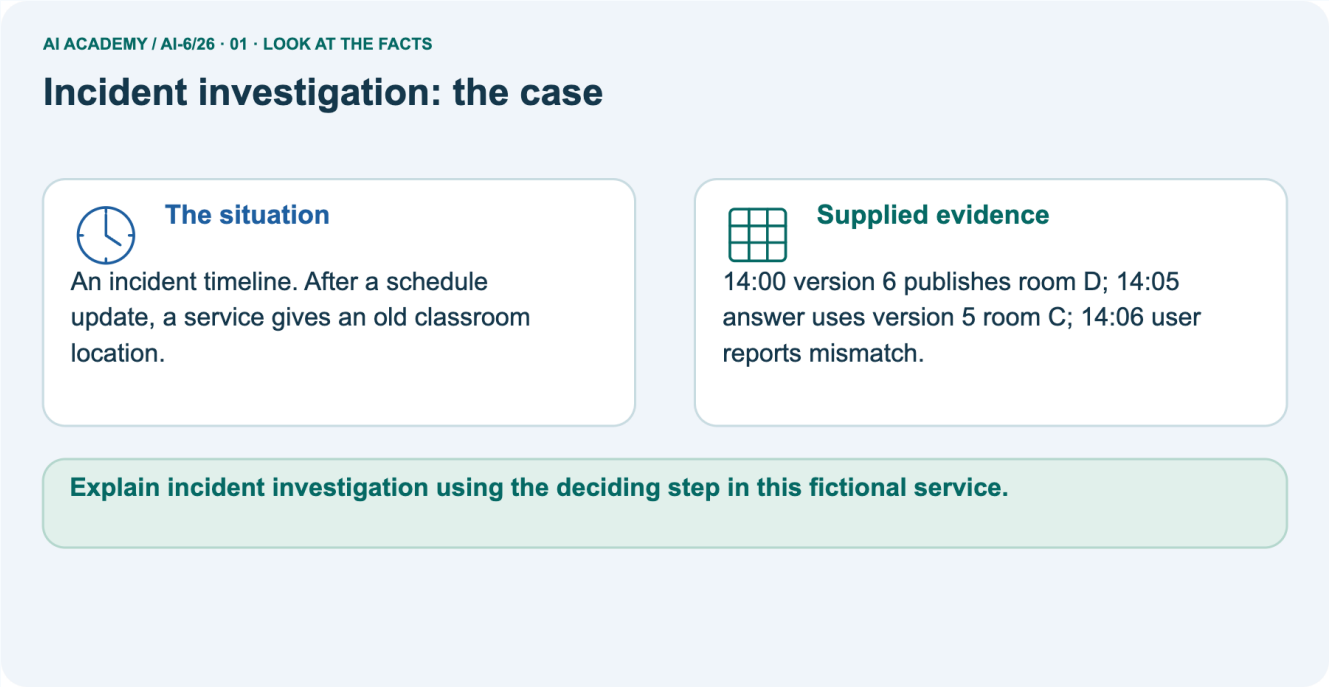
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

## Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

## 01. Incident investigation: the case

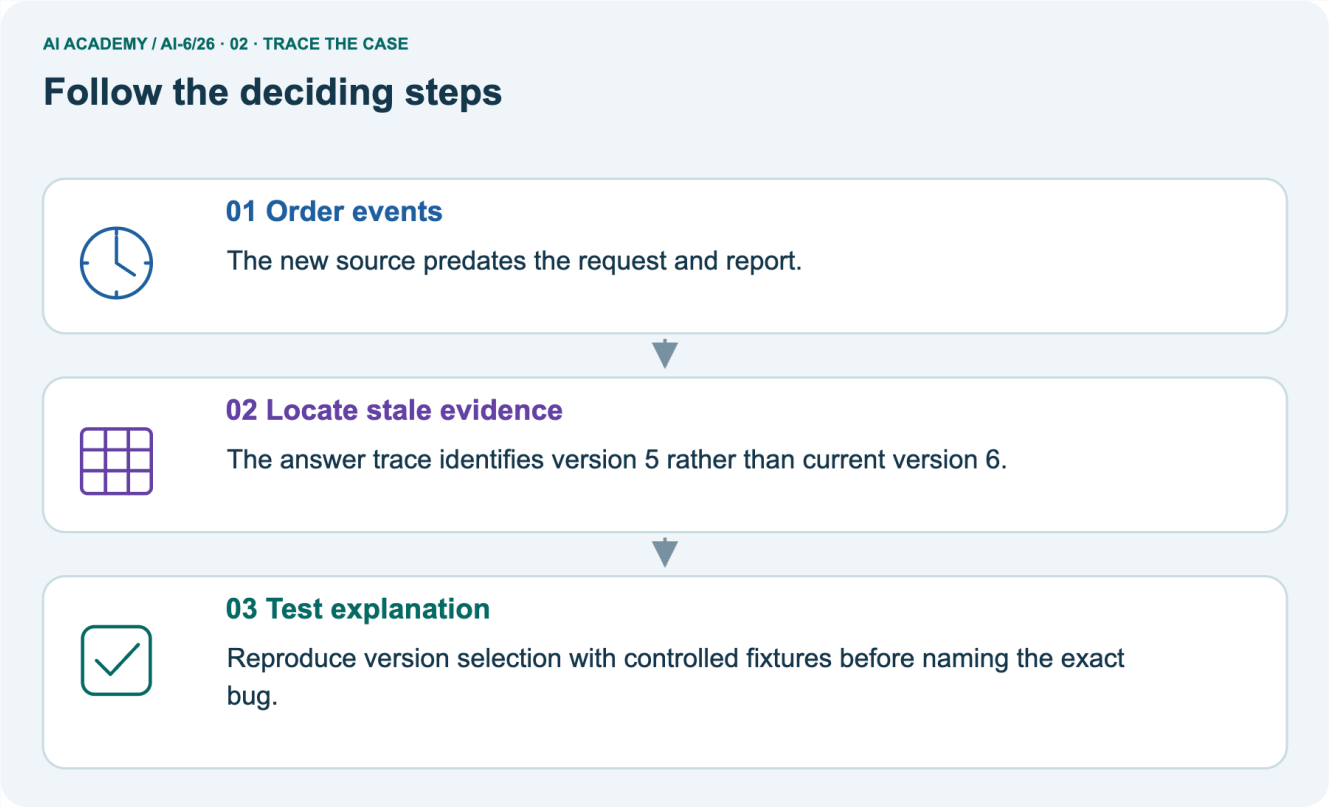



Follow the labelled picture with the matching explanation. Open the vector diagram

An **incident timeline**. After a schedule update, a service gives an old classroom location. Supplied facts: 14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Order events: The new source predates the request and report. → Locate stale **evidence**: The answer trace identifies version 5 rather than current version 6. → Test explanation: Reproduce version selection with controlled fixtures before naming the exact bug.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. Incident investigation — evidence trace

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Incident investigation — evidence trace

| Input and scope | Check or calculation | Conclusion and limit |
| --- | --- | --- |
| At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against evidence. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction. |

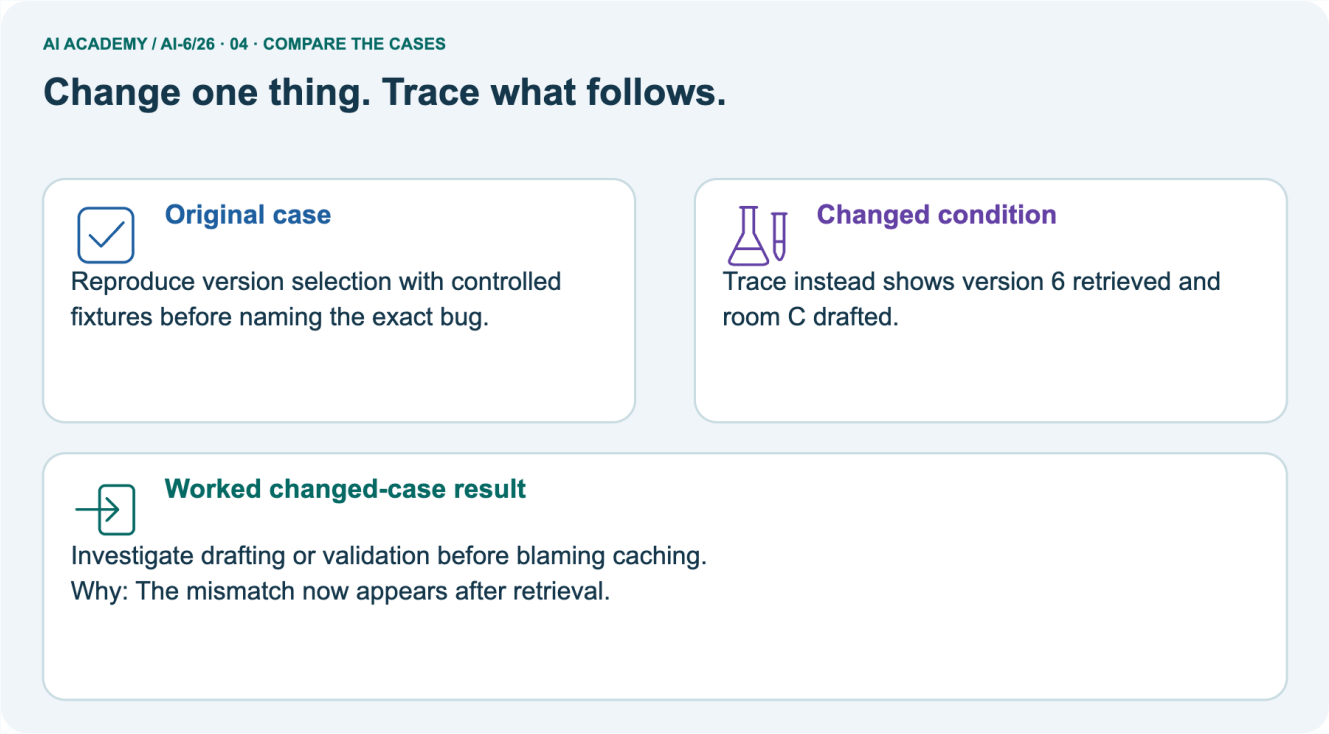
The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Incident investigation — **evidence** trace. At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



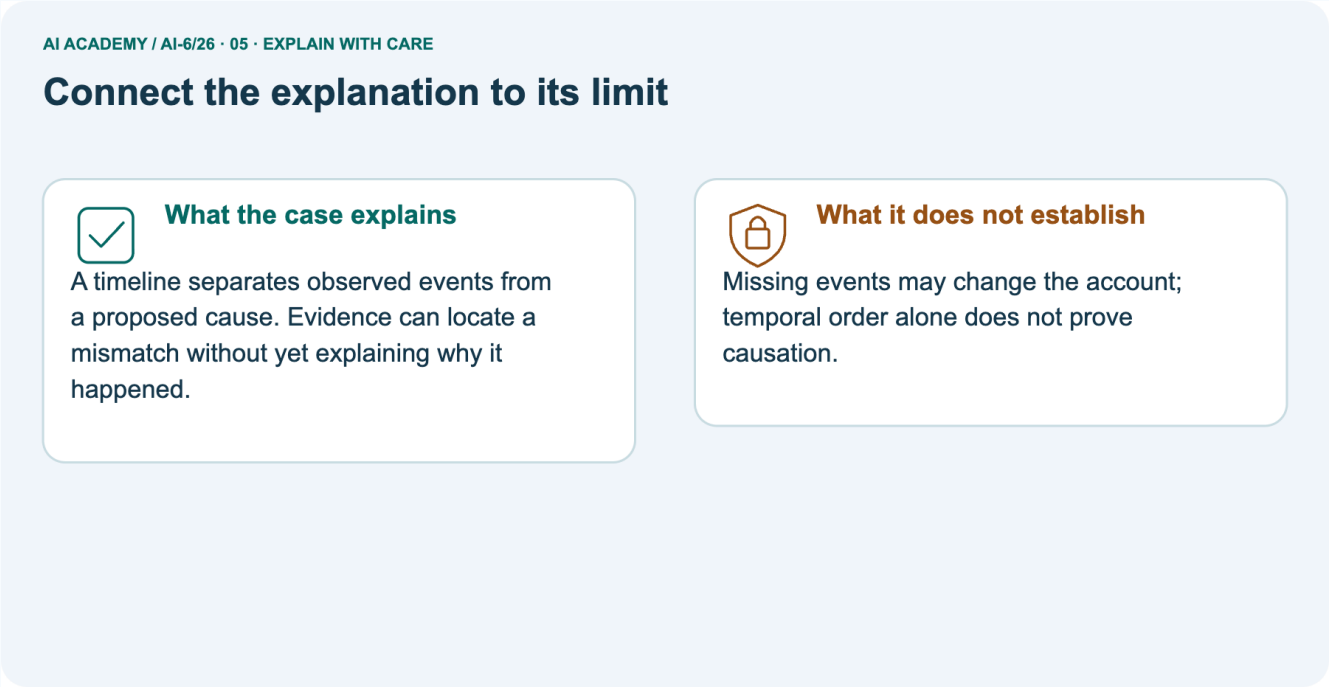

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reproduce version selection with controlled fixtures before naming the exact bug. Changed condition: Trace instead shows version 6 retrieved and room C drafted.

Expected result: Investigate drafting or validation before blaming caching. Reason: The mismatch now appears after retrieval.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit

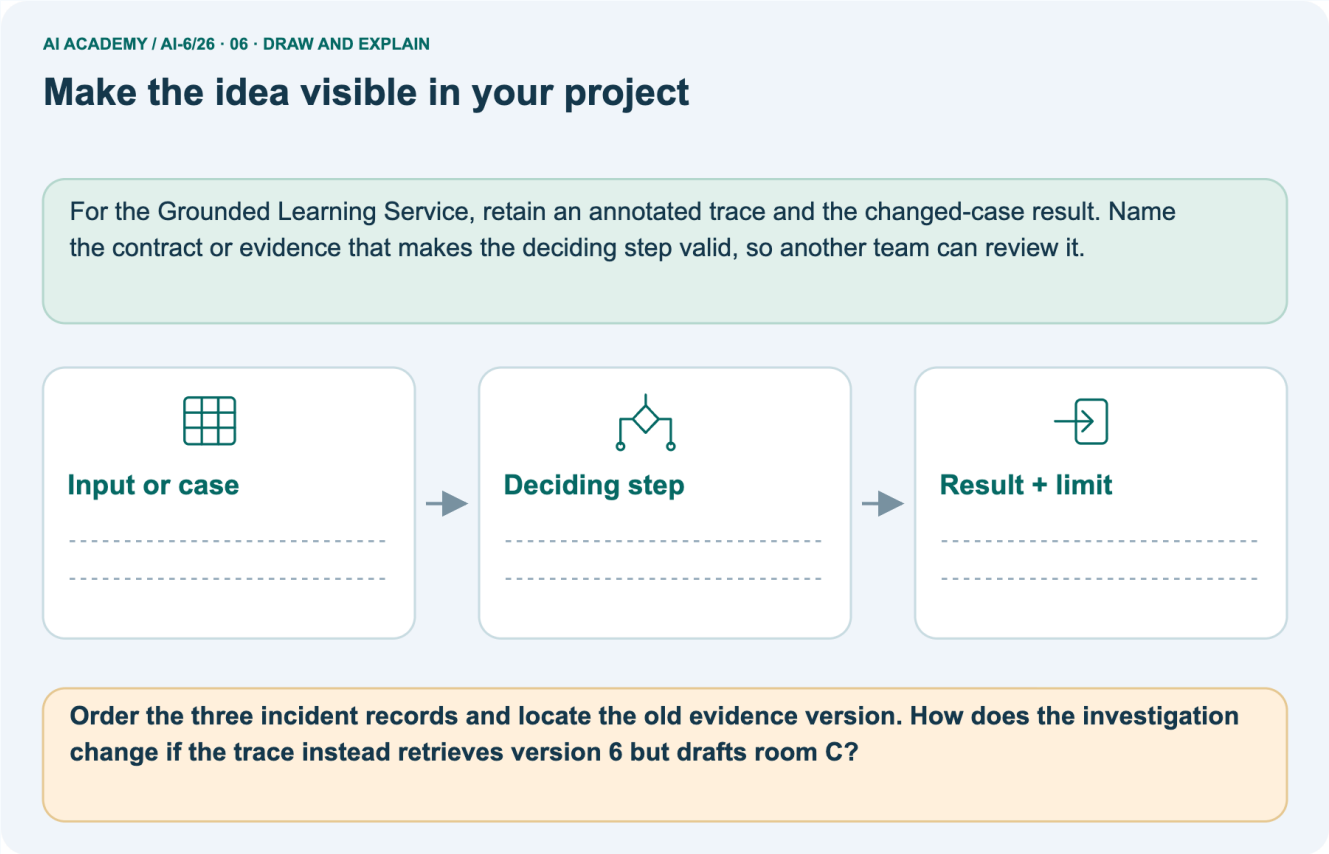



Follow the labelled picture with the matching explanation. Open the vector diagram

A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened. Limit: Missing events may change the account; temporal order alone does not prove causation.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

**01. Incident investigation: the case**

**02. Follow the deciding steps**

**03. Incident investigation — evidence trace**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.